CS207 Design and Analysis of Algorithms
Quiz

IIT Guwahati
Dept. of CSE
April 19, 2022

Roll No: | **Name:**

Q1. Three weights of 10, 20 and 30 kg have values of 60, 100 and 120 Rupees respectively. If the knapsack has a carrying capacity of 50 kg, then the optimal solutions of (a) the Fractional knapsack problem, and (b) the 0-1 knapsack problem are: (Show your work.) [4 marks]

(a). FKS considers the items in a nondecreasing order of value per weight. So 1, 2, 3 is the order. Items 1, 2 and $\frac{2}{3}$rds of 3 are picked. Optimal value $= 60 + 100 + \frac{2}{3} \times 120$
$= 240$ Rs.

(b). The subsets have the following values
000 → 0   001 → 120   010 → 100
011 → 220   100 → 60   101 → 180
110 → 160   111 – 280.
111 exceeds 50 kg. Not feasible
011 is the optimal soln
opt. value = 100 + 120 = 220

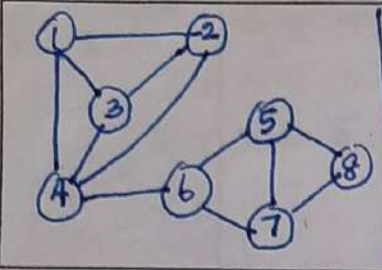
Q2. An algorithm performs a Inserts, a Extract-Mins and b Decrease-Keys on an initially empty priority queue. What is the time complexity of the algorithm, if the priority queue is implemented using (a) binary-heap (b) Fibonacci heap? [4 marks]

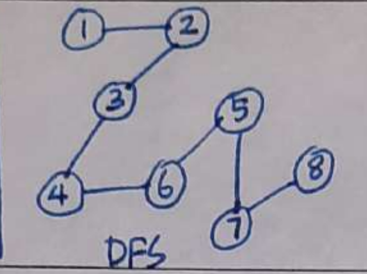
Any such mixture of operations would cause the heap to grow to a max. size of a. So a binary min heap would take $O((a+b)\log a)$ & a Fibonacci heap would take $O(a \log a + b)$ time on such a mixture

Q3. Explain why there are no forward nontree edges with respect to a BFS tree constructed for a directed graph. [4 marks]

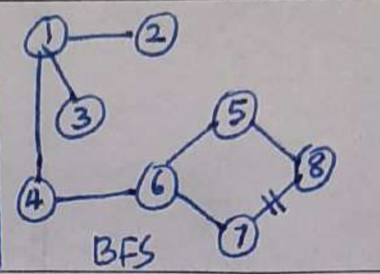
Suppose (u,v) is a nontree forward edge. So, v is a descendant of u; but is not a child of u; else, (u,v) would be a tree edge. So $p(v)$ is a descendant u. When u was visited, $p(v)$ was as yet unvisited, and so v was neither visited nor queued. So v ought to have been queued then, making u its parent.

Q4. Let G be a graph whose vertices are the integers 1 through 8, and let the adjacent vertices of each vertex be given as follows in the "vertex:(adjacent vertices)" format: 1:(2, 3, 4); 2:(1, 3, 4); 3:(1, 2, 4); 4:(1, 2, 3, 6); 5:(6, 7, 8); 6:(4, 5, 7); 7:(5, 6, 8); 8:(5, 7). Assume that, in a traversal of G, the adjacent vertices of a given vertex are returned in the same order as they are listed above. Then (a) Draw G. (b) Order the vertices as they are visited in a DFS traversal starting at vertex 1. (c) Order the vertices as they are visited in a BFS traversal starting at vertex 1. [4 marks]





1
2
3
4
6
5
7
8



1
2
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6
5
7
8

Q5. What is the best way to multiply a chain of matrices with dimensions that are 10×5, 5×2, 2×12, 12×60? Show your work. [4 marks]

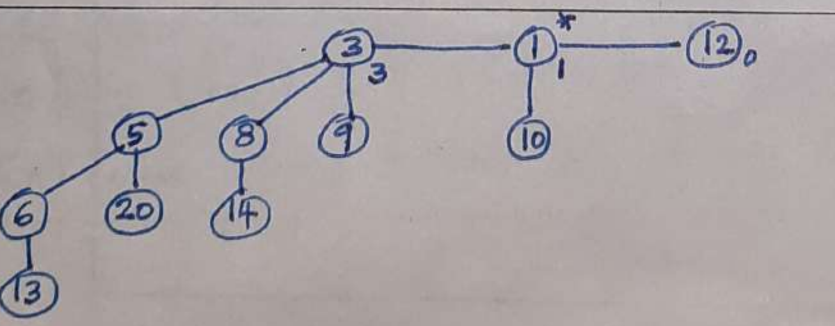
$$m_{13} = \min \begin{cases} m_{12} + 10 \times 2 \times 12 = 100 + 240 = 340 \\ m_{23} + 10 \times 5 \times 12 = 120 + 600 = 720 \end{cases} = 340$$

$$m_{24} = \min \begin{cases} m_{23} + 5 \times 12 \times 60 = 120 + 3600 = 3720 \\ m_{34} + 5 \times 2 \times 60 = 1440 + 600 = 2040 \end{cases} = 2040$$

$$m_{14} = \min \begin{cases} m_{13} + 10 \times 12 \times 60 = 340 + 7200 = 7540 \\ m_{12} + m_{34} + 10 \times 2 \times 60 = 100 + 1440 + 1200 = 2740 \\ m_{24} + 10 \times 5 \times 60 = 2040 + 3000 = 5040 \end{cases} = 2740$$

The best way is $(A_1 * A_2) * (A_3 * A_4)$

Q6. We have a binomial heap that was initially empty. Then 14, 8, 3, 9, 5, 20, 13, 6, 1, 10 and 12 were inserted into it in that order. Sketch the final heap. [4 marks]



*: min

Q7. Suppose there are n spears and n soldiers. Each soldier must be assigned a spear. The soldiers and spears are of various heights. Say, the i-th soldier has a height of h_i and her spear has a height of s_i. The assignment has to minimize $\sum_{i=1}^{n} |h_i - s_i|$. Consider a greedy strategy where we repeatedly pick from the as yet unassigned soldiers and spears the soldier-spear pair that has the least absolute difference in height. Prove or disprove that this greedy strategy results in the optimal assignment of spears to soldiers. [5 marks]

Consider 2 soldiers of heights h & $h+3$
Consider 2 spears of heights $h+1$ & $h-2$

Greedy pairings : $(h, h+1), (h+3, h-2)$
$\rightarrow 1+5 = 6$

The only other pairing : $(h, h-2), (h+3, h+1)$
$\rightarrow 2+2 = 4$

The greedy strategy doesn't give the optimal pairing

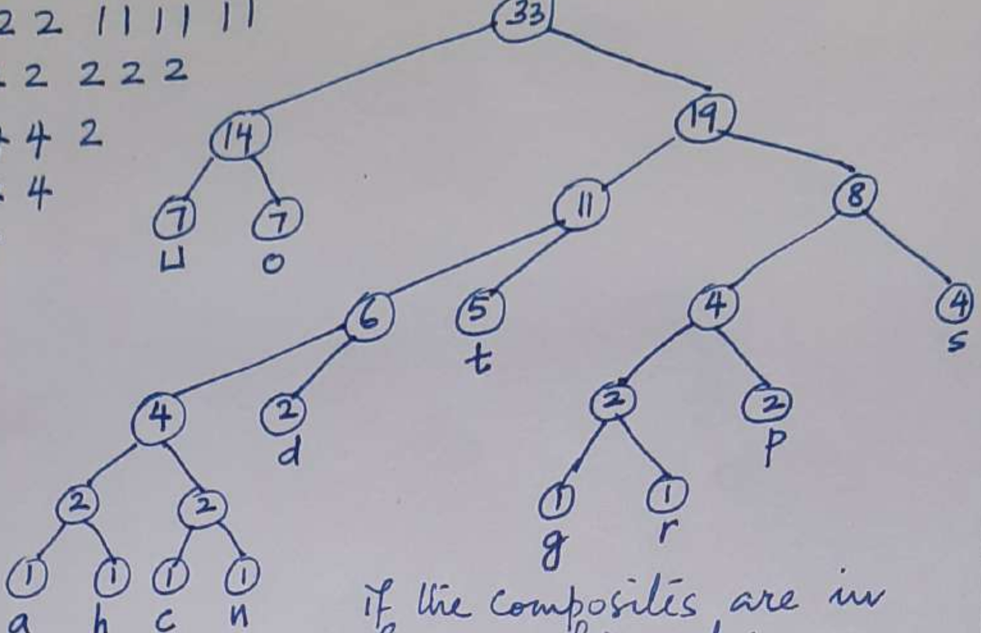
Q8. Suppose you are given an instance of the fractional knapsack problem in which all the items have the same weight. Show that you can solve the fractional knapsack problem in this case in $O(n)$ time. [8 marks]

W: the capacity of the knapsack, n: # items
x: the wt. of each item.
The soln. needs picking the $\lfloor W/x \rfloor$ costliest items, and a $(W - x\lfloor W/x \rfloor)/x$ fraction of the $(\lfloor W/x \rfloor + 1)$-st costliest item
This item can be found in $O(n)$ time using Blum's algo.
The costlier items can be marked in $O(n)$ time.
Overall time complexity : $O(n)$
(Blum's algo. finds the k^{th} smallest in an array of size n. for $1 \leq k \leq n$)

Q9. Draw the frequency table and Huffman tree for the following string: "dogs do not spot hot pots or cats". [5 marks]

␣	7
a	1
c	1
d	2
g	1
h	1
n	1
o	7
p	2
r	1
s	4
t	5

7 7 5 4 2 2 1 1 1 1 1 1
7 7 5 4 2 2 2 2 2
7 7 5 4 4 4 2
7 7 6 5 4 4
8 7 7 6 5
11 8 7 7
14 11 8
19 14
33



if the composites are in lexicographic ordering

Q10. Suppose you are given an integer c and an array, A, indexed from 1 to n, of n integers in the range from 1 to $5n$ (possibly with duplicates). Describe (in English, not in psuedocode) an efficient algorithm for determining if there are two integers, $A[i]$ and $A[j]$, in A that sum to c, so that $c = A[i] + A[j]$, for $1 \leq i < j \leq n$. What is the running time of your algorithm? [8 marks]

Initialize an array B of size 5n to all zero, and form the frequency distribution of A in it: $\forall i$, $1 \leq i \leq n$, B[A[i]]++;
For x, $1 \leq x \leq 5n$, if ($B[x] > 0$ and $1 \leq c-x \leq 5n$), check if $B[c-x] > 0$ if $c-x \neq x$, if $B[c-x] > 1$ if $c-x = x$.
The above takes $O(n)$ time.
Once A[i] and A[j] s.t. $A[i]+A[j] = c$ are found, one pair (i,j) s.t $1 \leq i < j \leq n$ can be found in $O(n)$ time using Linear Search in A.